Problem Formulation: Dopamine and Temporal-Difference Learning

Project 22 · Real-World RL (Unit 5 homework)

1. Problem

Dopamine neurons in animals respond to the surprise in a reward, not to the reward itself (Schultz, Dayan & Montague, 1997). This project asks whether a plain RL rule, tabular TD(0), reproduces three observed patterns: (1) before learning, the neurons fire at an unexpected food reward; (2) after learning, they fire at the predictive bell and no longer at the food; (3) if the food is omitted, activity dips below baseline at the expected time.

2. MDP formulation

Task type	Prediction (policy evaluation). The agent learns to predict reward; there is nothing to control.
Environment	One trial has 13 time steps (t = 0…12). A bell (cue) appears at t = 2 and food arrives at t = 8. The trial repeats for 300 trials (episodic).
State	The time step t within the trial. A table V(t) holds one value per step. The states before the bell are never updated, because the bell arrives at an unpredictable moment.
Action	None. The agent only observes (a single “wait” step with no effect on the environment).
Reward	r(t) = 1 when the food arrives (t = 8), otherwise 0.
Discount	γ = 1 (the task is episodic, so no discounting is needed).
Learned quantity	V(t): expected future reward from step t onward (initialised to 0).

3. Algorithm: TD(0)

TD error: $\delta(t) = r(t) + \gamma \cdot V(t) - V(t-1)$ (what happened minus what was expected)

Update: $V(t-1) \leftarrow V(t-1) + \alpha \cdot \delta(t)$ with learning rate α = 0.1

4. Evaluation and baseline

- Learning curve: δ at the bell and δ at the food, tracked over the 300 trials.
- Three probes with learning switched off: a trial before learning, one after learning, and one where the food is omitted.
- Baseline: an agent that never learns (V stays 0). Its surprise at the food stays at 1 on every trial.
- Success criteria: δ at the food → 0, δ at the bell → 1, and δ = −1 at the food time when the food is omitted. These are compared qualitatively with the published dopamine recordings.

5. Extra experiments (beyond the three required patterns)

- Heatmap of δ over time and trials, showing the surprise moving backward from the food to the bell.
- Learning-rate sweep, α = 0.03, 0.1, 0.3, 0.6.
- Reward probability from 0 to 1 (after Fiorillo, Tobler & Schultz, 2003).
- Timing change: food arriving 2 steps early or late (after Hollerman & Schultz, 1998).

6. Assumptions and simplifications

Fixed timing, one table entry per time step, no neural detail, and no real recordings. The comparison with the dopamine data is qualitative only, so the results show that TD error is a good explanation of the data, not that the brain computes it this way.